

Document Title: XR-500 Service Manual Product Name: XR-500 Industrial Cooling Controller
Document Type: Service Manual Revision Number: Rev 3.0 Document ID:
MAN-XR500-SVC-002 Date: April 5, 2025

1. **Product Overview** This manual provides technical service instructions, operational boundaries, and component replacement procedures for field technicians servicing the XR-500 Industrial Cooling Controller. The XR-500 controls industrial fluid chillers, secondary loop distribution, and precision thermal equipment by managing electronic expansion valves, pumps, and fan banks. Housed in a die-cast aluminum chassis rated to IP65, the system is designed to perform reliably in demanding plant environments. Adherence to the technical parameters herein is required during all maintenance and repair activities.
2. **Technical Specifications**
 - Operating Power Supply: 24V DC nominal (Permitted supply range: 21.6V DC to 26.4V DC)
 - Power Rating: 48 Watts maximum (2.0 A)
 - Temperature Inputs: 4 Channels, PT1000 RTD inputs (3-wire)
 - Control Outputs: 4x Analog 0–10V / 4–20 mA outputs
 - Auxiliary Relays: 4 Form C relays, rated for 250V AC, 5A (resistive)
 - Fluid System Maximum Pressure Rating: 1.8 MPa (18 bar)
 - Flow Rate Range: 50 L/min to 200 L/min (Nominal operating point: 130 L/min)
 - Communications Protocols: RS-485 (Modbus RTU), BACnet/IP, Modbus TCP
 - Ambient Temperature Range: -10°C to 55°C
 - Ingress Protection Standard: IP65
 - Enclosure Dimensions: 240 mm x 180 mm x 90 mm
 - Total Weight: 3.5 kg
3. **Installation Requirements** When re-mounting the unit following field servicing, attach the enclosure to a flat surface rated for a minimum load of 20 kg. Fasten securely with four M6 stainless steel bolts torqued to 4.5 N·m. Retain a minimum 150 mm open space surrounding all sides of the box to allow effective passive heat dissipation. All field wiring entering the bottom of the enclosure must utilize IP65 cable glands tightened to 3.5 N·m torque. Maintain strict physical separation between low-voltage signal wires (RTD inputs, 0–10V outputs, Ethernet, RS-485) and AC relay power cables. RS-485 communication bus cabling requires 120-ohm shielded twisted pair, with the shield grounded solely at the main controller chassis ground connection point.
4. **Operating Conditions** The XR-500 is specified for continuous operation in environments between -10°C and 55°C ambient temperature, with non-condensing relative humidity between 5% and 95%. For operating locations above 2000 meters elevation, derate the maximum operating ambient temperature by 0.5°C per 100 meters. Protect the controller body from direct solar exposure to prevent unnecessary thermal load. Permissible coolants include propylene glycol mixtures up to 50% concentration or treated industrial process water. Safe operation requires that the maximum fluid system pressure never exceed 1.8 MPa under any load condition. Fluid loop flow rates below 50 L/min will generate an automatic low-flow alarm and cause system safety cutouts.

5. Safety Information Field service must be performed exclusively by qualified industrial automation or refrigeration maintenance personnel. High AC potential up to 250V AC may remain present across relay contacts even when the 24V DC power feed to the XR-500 is disconnected. Isolate and tag out all power sources before opening the main access panel. Ensure proper grounding procedures and antistatic grounding wrist straps are used when servicing internal printed circuit assemblies. Exceeding the maximum DC supply limit of 30V DC can cause severe damage to internal power control circuitry.
6. Maintenance / Service Procedures Routine Inspection Routine (Bi-Annual):
 - Inspect outer casing, cable entry glands, and rubber door seals. Replace compromised IP65 glands immediately.
 - Re-torque internal screw terminal connectors to 0.5 N·m to guard against vibration loosening.
 - Clean external housing using a lint-free cloth moistened with mild detergent. Annual Servicing:
 - Validate accuracy of all 4 PT1000 RTD temperature channels using a calibrated precision resistance standard.
 - Verify system pressure sensor calibration to ensure the hydronic line does not exceed its maximum operating limit of 1.8 MPa.
 - Test internal CR2032 real-time clock backup battery; replace if cell voltage drops below 2.8V DC.
7. Troubleshooting
 - Symptom: Unit unresponsive / LEDs dark. Possible Cause: Loss of 24V DC input supply or reversed line polarity. Correction: Check source supply. Confirm voltage at input terminals measures between 21.6V DC and 26.4V DC. Verify correct wiring orientation.
 - Symptom: Fault Code E-102 (System High Pressure). Possible Cause: Controlled liquid loop pressure exceeding the maximum 1.8 MPa structural rating. Correction: Reduce loop pressure immediately. Check safety relief valves, inline expansion vessels, and system pressure transducer outputs.
 - Symptom: Fault Code E-204 (Low Flow Alarm). Possible Cause: Circulation fluid rate falling below 50 L/min limit. Correction: Inspect strainers, flush lines, verify pump control signals, and return system flow to nominal 130 L/min rate.
 - Symptom: RS-485 network communication failures. Possible Cause: Incorrect bus termination or multiple shield ground points. Correction: Set 120-ohm termination resistor switches only on the end physical nodes of the loop. Verify shield ground is attached at one point only.
8. Additional Notes To perform field firmware updates, connect a service PC to the primary Ethernet port and open the XR-500 Service & Configuration Tool (v3.2 or later). Always save user parameter profiles to local storage prior to initiating firmware programming. The onboard real-time clock (RTC) battery (CR2032) provides parameter memory backup during power loss, with a standard operating lifespan of 5 years.

8.